
Supporting information

AI as an Amplifier in Chemistry Laboratories: A SERG-Based Framework for Embodied and Assessable Instruction

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Appendix S5 (SI): Task Sheet, Micro-Lesson Design Card, and Unit/Project Blueprint (Blank Templates)

Note. These templates can be copied directly. Fields use **【 】** as placeholders. The three templates target: student execution (**task sheet**), a 15–20-minute teacher-led segment (**micro-lesson design card**), and a unit/project level plan (**lesson blueprint**). Each embeds **fallback**, **evidence chain**, and **compliance** elements to address the three problems: **misalignment** / **instability** / **missing evaluation**.

S5.1. Student Task Sheet (Blank)

Course/Class: **【 】** **Instructor:** **【 】** **Date:** **【 】**

Lab Topic: **【 】** (e.g., acid–base titration endpoint / HPLC calibration & quantification / kinetics fitting)

Learning Objectives (observable):

- Knowledge: **【 】** Skills: **【 】** Evidence expression (incl. uncertainty): **【 】**

1. Task Requirements

1. **Prepare:** Read the guide; obtain/download the data template (**【HPLC/ Titration/ Kinetics】**).
2. **Execute:** Collect/enter data as instructed → run Notebook or Excel fallback → export figures and parameters.
3. **Communicate:** In the report, provide your conclusion **plus** uncertainty/error sources (100–150 words).
4. **Submit:** Upload **【raw data + figures + report】** to **【class folder/platform】** ; complete self- and peer-assessment.

2. Step Checklist (tick when done)

- ☐ Obtain the data template and name files as required.
- ☐ Collect/enter data (recorder: **【 】**).
- ☐ Run analysis (import → clean → fit → visualize → export).
- ☐ Mark outlier handling and uncertainty estimation in the report.
- ☐ One self-assessment + at least one peer-assessment.
- ☐ Final submission and filename/format check.

3. Data Records

- **Data filename:** **【 】** **Collection date:** **【 】** **Operator:** **【 】**
- **Key parameters/settings:** **【 】** (e.g., wavelength/flow rate/indicator/temperature)

4. Results & Uncertainty (brief table)

Indicator	Value / Figure File	Uncertainty / Error Source	Impact & Improvement
【Concentration / Endpoint volume / Rate constant ...】	【 】	【 】	【 】

5. Self- & Peer-Assessment (excerpt from rubric)

- Is our **evidence chain** complete (data → analysis → conclusion, incl. uncertainty)? **【Yes/No, evidence:】**
- Are our **figures** conformant (units/labels/readable trends)? **【Yes/No, evidence:】**
- Peer key points (advice we received): **【】**

6. Fallback & Compliance (read first)

- **Fallback paths:** Script ↔ Excel; Local ↔ Cloud; Pre-generated figures. If a failure occurs, **switch within 5 minutes**.
- **Compliance tips:** Do not upload identifiable images; de-identify filenames/data; follow lab safety rules.

S5.2. Micro-Lesson Design Card (Blank, 15–20 minutes)

Course/Class: **【】** Teacher: **【】** Segment: 15–20 minutes (embedded)

Topic: **【】** SAMR target tier: S / A / M / R (check)

TPACK alignment (one sentence): Technology **【】** serves disciplinary objective **【】** via pedagogy **【】**.

1. Observable Learning Objectives (LOs)

1. **【】** 2) **【】** 3) **【】**

2. Teaching Flow (suggested template)

Time	Teacher Activity	Student Activity	AI Touchpoint	Evidence Output	Fallback
3' Intro	State goals/context	Respond/predict	—	Exit ticket	Mini-quiz (pre-made)
10' Execute	Explain the minimal data loop	Enter/run/plot	Notebook / Excel	Raw table + figure	Excel template / local script
5' Discuss	Prompt & coach	Peer review	Visualization board	Peer-review form	Oral brief
2' Wrap-up	Revisit error & conclusion	Submit & self-assess	—	Self-assessment slip	—

3. Resources & Equipment

- **Data template:** **【】** **Notebook/Excel files:** **【】** **Display/Camera:** **【】**
- **Accounts & permissions:** **【】** (offline pack / pre-generated figures path)

4. Assessment Design (formative-first)

- **Rubric indicators:** Data quality / Uncertainty / Conclusion consistency / Reflection
- **Evidence capture:** Task sheet, self/peer forms, exported figures & parameters

5. Risks & Compliance

- **RAMP:** Hazard **【】** Risk **【】** Control **【】** Emergency **【】**
- **Privacy:** De-identification & consent notice **【】**

- **5-minute switch plan:** Cloud → Local; Script → Excel; Camera → Pre-generated figures

65 S5.3. Lesson Blueprint (Blank, Unit/Project Level)

Course Title: 【】 **Grade/Contact hours:** 【】 **Lead/Co-teachers:** 【】
Unit Theme: 【】 **Public output for audience:** 【Poster / Defense / Brief / Video】

1. Driving Question

【Example: How can HPLC be used to verify whether the solubility of a drug meets the specification?】

70 2. Learning Objectives (observable)

- **Knowledge:** 【】 **Skills:** 【】 **Evidence & literacy (incl. uncertainty):** 【】

3. Milestones

Milestone	Key Activities	Deliverables	Time
M1	Propose plan / team roles	Draft plan	【】
M2	Collect & analyze	Data & figures	【】
M3	Review & revise	Version log	【】
M4	Public presentation	Final artifact + defense	【】

4. Learning Activities & AI Touchpoints (TPACK × SAMR alignment)

Activity	Disciplinary Points	Pedagogy /Strategy	AI Touchpoint	Evidence Capture	UDL Scaffolds / Fallback
【】	【】	【】	Collect / Analyze / Feedback / Evaluate	Task sheet / Rubric	Notebook ↔ Excel; Local ↔ Cloud

5. Assessment Plan (formative + summative)

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- **Rubrics:** Data quality, model choice & interpretation, visualization & communication, collaboration, uncertainty, compliance
 - **Evidence chain:** Raw data → processing script/table → figures → conclusions → reflection & improvement
 - **Scoring & audit:** Two raters; κ log; discrepancy adjudication

6. Resources & Supports (UDL 3.0)

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- **Multiple representations:** Handout / short video / sample datasets / exemplar reports
 - **Multiple engagements:** Roles (data, QA, presentation, audit)
 - **Multiple expressions:** Oral / visual / written; flexible language/format

7. Safety & Compliance (RAMP + Privacy)

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- **Hazard identification & risk assessment:** 【】 **Controls & PPE:** 【】 **Emergency procedures:** 【】
 - **Data governance:** Minimal capture, de-identified storage, consent notice, public-release checklist path: 【】

8. Timeline & Roles (Gantt/Task Table)

Task	Owner	Due	Dependencies	Status
【】	【】	【】	【】	In progress / Done

9. Risks & Alternatives

- **Risk categories:** Network / Devices / Materials / Time with **5-minute switch** routes
- **Pre-generated figures / offline data pack link:** 【】

90 10. Review & Transfer

- **Classroom data:** Outcome indicators & issue log
- **Redesign:** What to keep / drop / improve with rationale
- **Transfer points:** Feasible adaptations across classes/grades/disciplines

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